

# Scenario 1 - Level 2 (Ideal / Basic) — Fully Self-Funded

## Scenario Summary

Moderate-constraint path. All capital comes from founders. Team operates at 100% of ideal utilization (~480 productive technical hrs/mo). Phases run at midpoint durations with enough concurrency for parallel build and operating prep. No investor dilution, but founders must deploy significant personal capital. Operational and market risks remain the primary threats.

- Economic cost to Production: **\$175,000 / INR 14,525,000**
- Target timeline to Production: **24 months**
- Founder capital required: **\$175,000 / INR 14,525,000**
- Investor capital: **\$0**
- Equity: 60 / 20 / 0 (20% investor pool reserved, unissued)
- Main risk to watch: founder capital depletion before revenue ramp covers burn; no external safety net if costs overrun

## Timeline

| Phase                | Timing    | Duration | Minor Checkpoints                                                                                                                                                                                                                                                                  | Gate to Advance                                                                                               |
|----------------------|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Pre-Pilot            | M0 - M5   | 5 months | M1: Architecture freeze, issue triage complete. M2-M3: APK backlog locked, doctor training outline, web hardening parallel. M4: Internal rehearsals, callback SOP draft, admin runbooks started. M5: Full internal test cycles, admin runbooks versioned, APK core build underway. | Core web flows stable; internal rehearsals pass; doctor and admin runbooks versioned; APK build at 80%+.      |
| Pilot 1              | M5 - M8   | 3 months | M6: Warm-network onboarding begins, first subsidized consults. M7: Callback calibration, doctor playbook v1 live, throughput tracking active. M8: Throughput measured, escalation paths tested, Pilot 1 metrics compiled.                                                          | Ethiopia pilot runs live for 3 months with measured throughput, callbacks, and escalation handling.           |
| Pilot 2              | M8 - M16  | 8 months | M9-M10: Marketing push, user growth acceleration. M11-M12: Kenya prep, partner reporting active, APK hardening. M13-M14: Roster planning, partial automation, staffing calibration. M15-M16: Refresher training, reporting cleanup, compliance review.                             | 5k-10k user-capable operations with stronger APK, cleaner reporting, and staffed callback coverage.           |
| Production Readiness | M16 - M24 | 8 months | M17-M18: Compliance closeout, security hardening. M19-M20: Release freeze, launch playbooks drafted. M21-M22: Certification refreshers, production support schedule. M23-M24: Final QA cycles, launch readiness review.                                                            | Current prototype scope is production-safe across web and patient mobile with documented fallback operations. |

## Feature and Output Checklist

| Phase                | Output Checklist                                                                                                                                                                             |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre-Pilot            | Web core hardened (parallel tracks active); APK build at 80%+; doctor/admin training materials drafted; support desk scripts and callback trees defined; partner onboarding kits outlined.   |
| Pilot 1              | Patient app covers core journeys; provider web is stable for GP flow; doctor roster active; admin callbacks functional with SOP coverage; manual fulfilment visible in-system.               |
| Pilot 2              | Core patient features complete; partner dashboards functional; staffing ladder active; reporting, payments, and support loops tightened; Kenya market research complete; APK hardened.       |
| Production Readiness | Current prototype scope is production-grade across web and patient mobile; trained ops staff with manual fallback available; compliance documentation complete; release governance in place. |

## Services and SLAs

| Phase      | Uptime Target | Response Commitment            | Operating Model                                                                                                      |
|------------|---------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Pre-Pilot  | 96.0%         | Best-effort, same-day          | 3 core team + 7 part-time developers + 2 advisors; no employed clinical coverage; training and rehearsals only.      |
| Pilot 1    | 97.0%         | Same business day for critical | Minimum employed coverage: 2 doctors, 2 admin ops, 2 technical support. Partner clinics supplement.                  |
| Pilot 2    | 98.0%         | 4-hour P1 response             | Minimum employed coverage: 4 doctors, 4 admin ops, 2 technical support. Callback and partner reporting loops active. |
| Production | 99.0%         | 2-hour P1 response             | Minimum employed coverage: 6 doctors, 6 admin ops, 3 technical support for near-24/7 coverage.                       |

## Cost Breakdown

### Summary

| Category                                       | Amount                            |
|------------------------------------------------|-----------------------------------|
| Capital expenditure through Production         | \$88,400 / INR 7,337,200          |
| Operating expenditure before Production launch | \$86,600 / INR 7,187,800          |
| Total economic spend to Production             | <b>\$175,000 / INR 14,525,000</b> |

Workstream Detail

| Workstream                              | Economic Cost (USD) | Economic Cost (INR) | Notes                                                                                                                                 |
|-----------------------------------------|---------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Product and founder office              | \$126,720           | INR 10,517,760      | 3 core team at \$20/hr x 88 hrs/mo x 24 months. Shorter timeline than L1 reduces total founder-month cost.                            |
| Web platform hardening                  | \$14,900            | INR 1,236,700       | 745 hrs at \$20/hr senior rate. Same scope as L1, compressed into fewer calendar months.                                              |
| Patient APK build                       | \$7,200             | INR 597,600         | Fixed quote: 480 hrs at \$15/hr blended.                                                                                              |
| QA, DevOps, security                    | \$5,500             | INR 456,500         | ~275 hrs at \$20/hr. Adequate coverage at ideal resourcing level; moderate residual risk.                                             |
| Admin and care operations               | \$82,090            | INR 6,813,470       | Ops staff ladder runs 19 months (Pilot 1 starts M5). Pilot 1: \$2,482/mo x 3, Pilot 2: \$4,361/mo x 8, Production: \$6,542/mo x 8.    |
| Partner onboarding and provider success | \$11,250            | INR 933,750         | 750 hrs parallel workstreams at \$15/hr blended. Same effort, tighter calendar.                                                       |
| Legal, accounting, compliance           | \$6,500             | INR 539,500         | Shorter timeline = fewer compliance touch-points than L1. Friendly-network pricing at economic value.                                 |
| Infrastructure, phones, and tools       | \$22,340            | INR 1,854,220       | Monthly infra costs accumulate over 24 months. Pre-Pilot: ~\$200/mo, Pilot 1: ~\$395/mo, Pilot 2: ~\$690/mo, Production: ~\$1,150/mo. |
| Marketing, travel, rehearsals           | \$16,000            | INR 1,328,000       | Concentrated across 24 months. Fewer travel cycles needed than L1 due to tighter timeline.                                            |

Effort and Team

| Function               | Staffing Pattern                                                           | Notes                                                                                            |
|------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Core team              | 3 part-time members @ <del>4 hrs/day each</del> (264 hrs/mo combined)      | Single owner/founder + 2 founding members. Time split across product, ops, and fundraising prep. |
| Technical delivery     | 7 part-time developers, modeled at <b>~480 productive technical hrs/mo</b> | Ideal utilization. Android and web tracks run in parallel with sufficient concurrency.           |
| Advisory board         | 2 fractional advisors (~20-30 hrs/mo combined)                             | Finance, compliance, and strategic review.                                                       |
| Employed care coverage | Pre-Pilot: 0 / Pilot 1: 2 doctors / Pilot 2: 4 / Production Readiness: 6   | INR 60,000/mo per doctor. Partner clinics supplement but minimum employed coverage is explicit.  |
| Employed admin ops     | Pre-Pilot: 0 / Pilot 1: 2 admins / Pilot 2: 4 / Production Readiness: 6    | INR 18,000/mo per admin. Callbacks, exception handling, manual fulfilment.                       |
| Technical support      | Pre-Pilot: 0 / Pilot 1: 2 / Pilot 2: 2 / Production Readiness: 3           | INR 25,000/mo per tech support staff.                                                            |

Revenue and Pricing

Revenue Sources by Phase

| Revenue Source           | Pilot 1 (M5-M8)                                | Pilot 2 (M8-M16)                         | Production Readiness (M16-M24)   |
|--------------------------|------------------------------------------------|------------------------------------------|----------------------------------|
| GP consults              | Subsidized, Ethiopia-first, ~\$2-3/interaction | Full Ethiopia + Kenya pricing tests      | Core revenue line                |
| Specialist consults      | Minimal                                        | Grows after referral throughput improves | Important margin contributor     |
| Pharmacy commission      | Small, admin-assisted                          | Repeatable, still manual-heavy           | Stronger with partner conversion |
| Diagnostics commission   | Small, admin-assisted                          | Better once order flow consistent        | Retention lever                  |
| Travel/care coordination | Few cases, high touch                          | Higher-value but low-volume              | Strategic differentiator         |

Pricing Benchmarks

| Revenue Line             | Ethiopia Benchmark | Kenya Benchmark     | Notes                                                          |
|--------------------------|--------------------|---------------------|----------------------------------------------------------------|
| GP consult               | \$3.50 / ETB 200   | \$5.00 / KES 650    | Starts subsidized in Pilot 1; reaches benchmark by Production. |
| Specialist consult       | \$9.00 / ETB 513   | \$12.00 / KES 1,560 | Pilot 2 onward.                                                |
| Pharmacy take-rate       | 10% of order value | 10% of order value  | Pilot 1 onward with manual fulfilment.                         |
| Diagnostics take-rate    | 12% of order value | 12% of order value  | Pilot 1 onward with admin/lab coordination.                    |
| Travel/care coordination | \$10-15 per case   | \$10-15 per case    | Small volume, high-touch service.                              |
| Subscription (premium)   | Deferred           | Deferred            | Not modeled before meaningful scale.                           |

Cash Flow and Runway

Monthly Burn by Phase

| Phase                          | Monthly Team Burn | Monthly Ops Staff | Monthly Infra | Monthly Total Burn | Revenue Offset    |
|--------------------------------|-------------------|-------------------|---------------|--------------------|-------------------|
| Pre-Pilot (M0-M5)              | \$12,520          | \$0               | \$200         | \$12,720           | \$0               |
| Pilot 1 (M5-M8)                | \$12,520          | \$2,482           | \$395         | \$15,397           | \$200-500/mo      |
| Pilot 2 (M8-M16)               | \$12,520          | \$4,361           | \$690         | \$17,571           | \$500-3,000/mo    |
| Production Readiness (M16-M24) | \$12,520          | \$6,542           | \$1,150       | \$20,212           | \$3,000-15,000/mo |

### Monthly Cash Flow

| Month | Phase      | Founder Capital In | Revenue In | Outflow  | Net       | Ending Cash |
|-------|------------|--------------------|------------|----------|-----------|-------------|
| M1    | Pre-Pilot  | \$14,000           | \$0        | \$12,720 | \$1,280   | \$1,280     |
| M2    | Pre-Pilot  | \$14,000           | \$0        | \$12,720 | \$1,280   | \$2,560     |
| M3    | Pre-Pilot  | \$14,000           | \$0        | \$12,720 | \$1,280   | \$3,840     |
| M4    | Pre-Pilot  | \$14,000           | \$0        | \$12,720 | \$1,280   | \$5,120     |
| M5    | Pre-Pilot  | \$14,000           | \$0        | \$12,720 | \$1,280   | \$6,400     |
| M6    | Pilot 1    | \$14,000           | \$200      | \$15,397 | -\$1,197  | \$5,203     |
| M7    | Pilot 1    | \$14,000           | \$350      | \$15,397 | -\$1,047  | \$4,156     |
| M8    | Pilot 1    | \$14,000           | \$500      | \$15,397 | -\$897    | \$3,259     |
| M9    | Pilot 2    | \$7,000            | \$700      | \$17,571 | -\$9,871  | -\$6,612    |
| M10   | Pilot 2    | \$7,000            | \$1,000    | \$17,571 | -\$9,571  | -\$16,183   |
| M11   | Pilot 2    | \$7,000            | \$1,300    | \$17,571 | -\$9,271  | -\$25,454   |
| M12   | Pilot 2    | \$7,000            | \$1,600    | \$17,571 | -\$8,971  | -\$34,425   |
| M13   | Pilot 2    | \$7,000            | \$2,000    | \$17,571 | -\$8,571  | -\$42,996   |
| M14   | Pilot 2    | \$7,000            | \$2,300    | \$17,571 | -\$8,271  | -\$51,267   |
| M15   | Pilot 2    | \$7,000            | \$2,600    | \$17,571 | -\$7,971  | -\$59,238   |
| M16   | Pilot 2    | \$7,000            | \$3,000    | \$17,571 | -\$7,571  | -\$66,809   |
| M17   | Production | \$5,000            | \$4,000    | \$20,212 | -\$11,212 | -\$78,021   |
| M18   | Production | \$5,000            | \$5,500    | \$20,212 | -\$9,712  | -\$87,733   |
| M19   | Production | \$5,000            | \$7,000    | \$20,212 | -\$8,212  | -\$95,945   |
| M20   | Production | \$5,000            | \$8,500    | \$20,212 | -\$6,712  | -\$102,657  |
| M21   | Production | \$5,000            | \$10,000   | \$20,212 | -\$5,212  | -\$107,869  |
| M22   | Production | \$5,000            | \$12,000   | \$20,212 | -\$3,212  | -\$111,081  |
| M23   | Production | \$5,000            | \$14,000   | \$20,212 | -\$1,212  | -\$112,293  |
| M24   | Production | \$5,000            | \$15,000   | \$20,212 | -\$212    | -\$112,505  |

**Cumulative founder capital deployed:** ~\$175,000 (\$14,000/mo x 8 months + \$7,000/mo x 8 months + \$5,000/mo x 8 months)

**WARNING:** Cash goes negative at M9. This path requires founders to front-load capital or maintain a large lump-sum reserve to cover the negative cash position through the scaling phases. The gap between founder injections and burn rate widens significantly during Pilot 2 and Production. Unlike S1-L1 where the gap is gradual, S1-L2 hits larger monthly deficits sooner because burn is higher — but the total timeline is 10 months shorter.

### ROI and Break-Even

| Metric                                               | Value                                                                                                          |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Break-even month (monthly revenue >= monthly burn)   | ~ <b>M26-M32</b> (revenue must reach ~\$20,000/mo to cover Production burn; trajectory suggests M28-M30 range) |
| Cumulative break-even (total revenue >= total spend) | ~ <b>M36-M42</b> (depends on post-Production revenue growth sustaining \$15,000-25,000/mo)                     |
| ROI at 12 months post-production (M36)               | <b>-20% to -10%</b> (still recovering capital but closer than L1)                                              |
| ROI at 24 months post-production (M48)               | <b>+15% to +35%</b> (positive and growing)                                                                     |
| ROI at 36 months post-production (M60)               | <b>+50% to +80%</b> (healthy if growth sustains)                                                               |

### Key Risks

| # | Risk                                               | Likelihood | Impact   | Mitigation                                                                                                                                                                                               |
|---|----------------------------------------------------|------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Founder capital depletion before break-even</b> | High       | Critical | Founders must have \$175,000+ available. Cash goes negative at M9; requires pre-committed reserves or lump-sum injections. Consider converting to S2/S3 if capital reserves fall below 3 months of burn. |

|   |                                                    |             |      |                                                                                                                                                                                        |
|---|----------------------------------------------------|-------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | No external validation or safety net               | High        | High | Without an investor, there is no external check on product-market fit assumptions. Build in explicit go/no-go gates at each phase transition.                                          |
| 3 | Founder burnout over 24 months                     | Medium-High | High | Shorter than L1's 34 months but still a long self-funded journey. Higher burn means more financial stress. Build in explicit rest cycles and consider the S2 pivot if fatigue sets in. |
| 4 | Partner and clinic attrition during extended pilot | Medium      | High | 8-month Pilot 2 still risks losing early partners. Keep engagement through regular check-ins and incremental value delivery.                                                           |
| 5 | Market window closure                              | Medium      | High | 24-month timeline is competitive but East African health-tech landscape is evolving. Monitor competitor moves and be ready to accelerate via scenario conversion.                      |

## Recommendation

**Consider if founders have capital.** S1-L2 is the viable self-funded path. The 24-month timeline is competitive, the team operates at ideal utilization, and the total cost is meaningful but not unreasonable for founders with sufficient personal reserves.

This path is viable if:

1. Founders have personal savings or committed capital exceeding \$200,000 (buffer above the \$175,000 modeled cost)
2. Founders are willing to absorb a negative cash position starting at M9 and deepening through M24
3. The team can maintain energy and focus over a 24-month self-funded journey without investor validation
4. There is a credible fallback plan to convert to S2 or S3 if the M9 cash trigger proves unsustainable

**Key advantage over S1-L1:** 10 months faster, higher team utilization enables parallel execution, and the higher burn rate is offset by a shorter accumulation period — resulting in less total cost than L1's stretched timeline might suggest.

**Key disadvantage vs. S2/S3 paths:** Founders bear 100% of financial risk with no external safety net. If the market does not respond as modeled, there is no investor capital buffer to absorb the shortfall.

All figures reference *params.md* v2. Currency conversion: 1 USD = 83 INR.